



OPEN Optimal power scheduling in real-time distribution systems using crow search algorithm for enhanced microgrid performance

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Microgrids (MGs) have gained significant attention over the past two decades due to their advantages in service reliability, easy integration of renewable energy sources, high efficiency, and enhanced power quality. In India, low-voltage side customers face significant challenges in terms of power supply continuity and voltage regulation. This paper presents a novel approach for optimal power scheduling in a microgrid, aiming to provide uninterrupted power supply with improved voltage regulation (VR). To address these challenges, a crow search algorithm is developed for effective load scheduling within the distribution system. The proposed method minimizes the total operating cost (TOC) and maximizes VR under varying loading conditions and distributed generation (DG) configurations. A case study in Tamil Nadu, India, is conducted using a microgrid composed of three distributed generation sources (DGs), modeled and simulated using the Electrical Transient Analyzer Program (ETAP) environment. The proposed approach is tested under three operational scenarios: grid-connected mode, islanded mode, and grid-connected mode with one DG outage. Results indicate that the crow search algorithm significantly optimizes load scheduling, leading to a substantial reduction in power loss and enhancement in voltage profiles across all scenarios. The islanded mode operation using the crow search algorithm demonstrates a remarkable reduction in TOC and maximizes voltage regulation compared to other modes. The main contributions of this work include: (1) developing a new meta-heuristic approach for power scheduling in microgrids using the crow search algorithm, (2) achieving optimal power flow and load scheduling to minimize TOC and improve VR, and (3) successfully implementing the proposed methodology in a real-time distribution system using ETAP. The findings showcase the effectiveness of the crow search algorithm in microgrid power management and its potential for application in other real-time power distribution systems.

Keywords Microgrid, Load scheduling, Crow search algorithm, Voltage regulation, Distributed generation, Total operating cost, Power management, ETAP simulation

Nowadays, medium and low-voltage power grids are increasingly being transformed into microgrids, owing to their numerous benefits, including improved reliability, reduced power losses, and enhanced economic performance. This transformation is driven by the growing need for more flexible, sustainable, and resilient power systems, especially as the integration of renewable energy sources becomes more widespread. Microgrids are small-scale, localized systems that incorporate various components, such as distributed generation (DG)

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units—like solar photovoltaic (PV) panels, wind turbines, and small hydro plants—along with energy storage systems (ESS) and different types of loads. These systems can operate in both grid-connected mode, where they interact with the main utility grid, and islanded mode, where they function independently, providing a reliable power supply during grid outages or disturbances¹. The integration of microgrids brings about unique changes in the operational processes and planning of distribution systems. These changes include advanced load scheduling techniques that optimize the usage of available resources, self-healing capabilities to automatically detect and address faults in the network, a reduction in system expansion costs through strategic placement of DGs and ESS, and an improved response to fluctuating power demands. However, despite these benefits, the implementation of microgrids also presents significant challenges. The most critical of these is achieving optimal power scheduling of distributed resources without violating system constraints, such as voltage limits and power flow regulations. Additionally, the intermittent and irregular nature of renewable energy generation poses further complexities, making it difficult to sustain power balance within the distribution system². Therefore, addressing these challenges remains a key focus in the successful operation and management of microgrids.

Microgrids play a vital role in achieving optimal power scheduling, a process that is crucial for enhancing the efficiency and reliability of modern power distribution systems. Optimal power scheduling focuses on optimizing specific objective functions, including minimizing power losses, reducing total operational costs, improving voltage regulation, and enhancing overall system reliability. This process involves the careful coordination and control of various DG units, energy storage systems, and different load types to ensure that power supply meets demand efficiently. The scheduling strategy is based on several key factors: the transmission capabilities of DGs, the flexibility and prioritization of loads, and the potential for power exchange between microgrids and the main utility grid³. To achieve effective power scheduling, the entire process must adhere to a range of technological solutions and be consistent with operational constraints. These constraints include power flow regulations, voltage limits, and the dynamic nature of load demands. Moreover, the integration of microgrids necessitates the use of advanced control and optimization algorithms to manage power exchanges effectively, respond to real-time changes in generation and load conditions, and maintain system stability. Therefore, the choice of technologies and the development of robust control mechanisms are critical to the successful implementation of optimal power scheduling in microgrids.

In the context of optimizing modern power systems, particularly with the increasing integration of renewable energy sources, advanced optimization algorithms have been instrumental in addressing complex operational challenges. Starting with Pandya et al.⁴ who introduced the Multi-Objective Snow Ablation Optimization Algorithm (MOSAO), this study addresses the security-constrained optimal power flow (OPF) problem by incorporating wind energy sources alongside FACTS (Flexible AC Transmission Systems) devices. The MOSAO algorithm enhances system stability by managing the inherent uncertainties of renewable energy sources, ensuring more reliable power flow and energy distribution⁴. This approach correlates closely with the work of Izci et al.⁵, who focused on improving voltage regulation in power systems. By utilizing a fractional-order PID plus double-derivative (FOPIDD²) controller integrated with the Mountain Gazelle Optimizer (MGO), Izci and colleagues significantly improved transient response times, thus enhancing voltage stability—a critical factor in renewable-integrated systems where voltage fluctuations are common⁵.

Expanding upon these control mechanisms, Premkumar et al.⁶ introduced an intelligent multi-objective optimization approach for hybrid power systems that operate with stochastic renewable energy sources and FACTS devices. Their approach aligns with the objectives of Pandya's MOSAO in terms of system reliability and stability but focuses more on the optimal control of hybrid power sources to balance power quality and grid reliability⁶. The work of Pandya et al. in a separate 2024 study builds upon this by employing the Multi-objective RIME algorithm for a techno-economic analysis of power systems, specifically focusing on security-constrained load dispatch and power flow. This study uniquely addresses the economic considerations alongside technical constraints, highlighting the importance of robust decision-making frameworks that incorporate uncertainties in hybrid power system operations⁷. Further contributions to this field are seen in the work of Fadheel et al.⁸, who addressed the challenge of frequency regulation in multi-microgrid systems. They developed a hybrid Sparrow Search Optimization algorithm to optimize fractional virtual inertia control, ensuring stable and reliable system operation even amidst the variability of renewable inputs⁸. This aligns well with the demand-side optimization presented by Manzoor et al.⁹, who introduced the Arithmetic Harris Hawks Optimization (AHHO) algorithm for smart grid management. AHHO emphasizes efficient energy consumption patterns, improving the integration of renewables within the grid and enhancing overall system efficiency—a goal that resonates with the operational stability efforts in earlier studies⁹. Finally, the optimization of grid-connected photovoltaic (PV) systems is seen in two studies by Altawil et al.¹⁰. In the first, they employed a fractional-order PI controller optimized by the Slap Swarm Algorithm to stabilize grid voltage in PV systems, addressing the frequent fluctuations caused by solar energy's variable nature¹⁰. In their second study, Altawil and team used the Ant-Lion Optimization Algorithm for managing hybrid renewable energy systems. This method ensures optimal energy management and utilization within grid-connected systems, thereby enhancing efficiency and stability¹¹. Together, these studies contribute to a cohesive understanding of how various optimization techniques—whether aimed at voltage regulation, frequency stabilization, load dispatch, or economic considerations—enhance the operational resilience of power systems with high renewable penetration. Through this progression of research, it is clear that these algorithms not only support the technical demands but also address the economic and reliability aspects essential for modern power systems.

A wide range of working models and technical methods for microgrid operation has been explored by numerous researchers. These models cover various aspects of microgrid management, including power generation, storage, load management, and economic optimization. One notable area of study is the integration of day-ahead scheduling in microgrids, which involves coordinating the operation of DG units, storage systems (SS), and demand-side response (DSR) mechanisms to reduce the overall cost of energy¹². Day-ahead scheduling

is particularly important because it allows microgrid operators to optimize the use of available resources in advance, taking into account forecasted generation and load patterns. Demand-Side Response (DSR) offers an opportunity for load-side consumers to actively participate in microgrid operations by adjusting their power consumption in response to supply conditions, price signals, or incentives^{13,14}. Implementing DSR requires the use of algorithms that rely on forecasted information about the capacity of DGs, the number of storage systems, electricity prices, and expected load demand. This approach not only helps in managing energy costs but also contributes to grid stability by balancing supply and demand. To accurately model and implement these operations in microgrids, researchers have utilized advanced programming techniques such as stochastic and robust programming. These techniques account for uncertainties in renewable energy generation, fluctuating demand, and market prices, allowing microgrids to be more adaptive and resilient to real-world variations¹⁵. By using these models, operators can better manage resources, reduce operational costs, and maintain a reliable power supply^{16,17}.

The motivating problem of combining power scheduling, DG placement, and consumer price setting has been explored in several studies, such as¹⁸. This integrated approach aims to address the complex challenge of optimizing both the technical and economic aspects of microgrid operation. Specifically, the objective is often to maximize social welfare by balancing the profit margins for energy producers and the cost-effectiveness for consumers. In the context of microgrid research, most studies have traditionally focused on unit commitment (UC) and optimal power flow (OPF) problems as separate entities. Unit commitment involves determining the schedule for power generation units to meet the expected load demand in a cost-efficient manner. While an optimal power flow focuses on the real-time distribution of electrical power within the grid to minimize losses and enhance system reliability. However, the separation of these problems can lead to suboptimal outcomes, as the interdependencies between generation scheduling and power flow dynamics are not fully addressed. Only a limited number of research works have attempted to combine UC and OPF problems. And, those that do often focus on conventional power distribution systems without incorporating the complexities introduced by modern elements such as SSs, DGs, and DSR mechanisms^{19,20}. The integration of these components adds layers of complexity to the problem, requiring advanced algorithms and models to simultaneously address operational, economic, and technical constraints. This highlights the need for more holistic approaches that can effectively incorporate these emerging technologies into microgrid power scheduling.

Recent studies, such as those in^{21,22}, have explored centralized mutual methods specifically designed for standalone microgrids. Centralized methods focus on managing the power allocation problem by overseeing the distribution of power within a well-organized microgrid network. These approaches aim to optimize resource utilization, ensuring that power generation, storage, and consumption are balanced efficiently. The centralized approach has also been applied to microgrids in^{23–25} where a single controlling entity manages power distribution and exchange within the microgrid. In addition, a “jump and shift” method was proposed to address the combined UC and OPF problems in systems incorporating both SSs and DG units²⁶. This method offers a strategic way to determine the optimal schedule for power generation and distribution, considering the presence of various energy sources. Moreover, this approach has been modified in recent research for scenarios involving multiple interconnected microgrids, enabling better coordination and power sharing among different microgrids^{27,28}. Various techniques were implemented in the microgrid for energy management studies are discussed in^{29–31}. However, despite the effectiveness of these methods, it is important to note that the current paper does not employ specific power management approaches in its problem formulation^{32,33}. Instead, it focuses on other aspects of microgrid operation, such as optimal power scheduling using advanced algorithms, without delving into the detailed strategies of power management. This distinction highlights the need for further exploration into integrating power management techniques with the proposed methodologies.

Microgrids are increasingly being adopted in modern power distribution systems due to their ability to integrate renewable energy sources, enhance reliability, and offer flexible power management. However, achieving optimal power scheduling within microgrid presents significant challenges. These include managing the uncertainties of renewable energy generation, meeting diverse and dynamic load demands, ensuring voltage regulation (VR), and minimizing total operating costs (TOC).

Existing research often focuses on specific aspects of microgrid operation, such as minimizing system costs, optimizing frequency security, or addressing uncertainties in renewable energy. As shown in Table 1, recent works have aimed at objectives like minimizing electricity consumption costs¹⁴, system costs²⁴, and optimizing frequency security²⁸. However, most approaches do not comprehensively address both minimization of TOC

Type	Ref. ⁶	Ref. ¹³	Ref. ¹⁶	Proposed method
Objective	Minimization of user's electricity consumption cost	Minimization of system cost	Optimize the frequency security	Minimization of TOC and maximization of VR
Power scheduling	Yes	Yes	Yes	Yes
Bus voltage (security constraints)	No	No	Yes	Yes
Renewable energy resources (uncertainty)	No	Yes	No	Yes
Power flow type	DC-OPF	DC-OPF	AC-OPF	AC-OPF

Table 1. A methodical comparison between the performance of recent works and this work.

and improving VR in a unified manner. Additionally, while many studies consider UC and OPF problems separately, few have explored their integration in the context of microgrids with DGs, ESS, and DSR.

Table 1 highlights a comparative analysis of recent research efforts and the proposed method. It reveals gaps in existing work, such as the lack of focus on bus voltage security constraints, proper modeling of uncertainties in renewable energy resources, and the use of more simplistic DC-OPF models instead of AC-OPF for power flow analysis. These limitations emphasize the need for a comprehensive power scheduling approach that can handle real-time operating conditions, uncertainties, and the complexities of both active and reactive power flow.

1. For power scheduling, many optimization techniques have been developed, such as gravitational search algorithm³⁴, ant colony algorithm³⁵, artificial bee colony search algorithm³⁶, Binary Backtracking Search Algorithm³⁷, and genetic algorithm³⁸. These algorithms got difficult calculation, boundaries, coding, complications, making, and wide computational time for the best optimum solution. Moreover, based on the literature, there are still limitations related to microgrid power scheduling. Therefore, a crow search algorithm is proposed in this study to overcome the limitation of the above mentioned algorithms.
2. This paper proposes a novel approach to address these research gaps. The main contributions of this work are as follows:
 - Developed a new meta-heuristic algorithm for optimal power scheduling of DG sources, focusing on minimizing total operating cost (TOC) and improving voltage regulation (VR).
 - Properly modeled uncertainties of renewable energy resources and diverse load types, allowing for robust and adaptable power management in microgrids.
 - Utilized AC Optimal Power Flow (AC-OPF) for power scheduling, providing a more accurate reflection of active and reactive power flow compared to DC-OPF.
 - Included bus voltage security constraints in the model, enhancing the reliability and stability of microgrid operations.
 - Conducted a comparative analysis with recent works, demonstrating the effectiveness of the proposed method in addressing gaps in power scheduling strategies.
 - Employed Electrical Transient Analysis Program (ETAP) to develop a detailed microgrid case study for continuous modeling and operation, showcasing the real-time applicability of the proposed method.

The remainder of this manuscript is organized as follows: Section “[Modeling of microgrid optimal scheduling](#)” provides a comprehensive review of related works and the challenges associated with microgrid power scheduling. Section “[Problem formulation](#)” presents the detailed problem formulation, including the objectives and constraints for optimal power scheduling in microgrids. Section “[Crow search algorithm](#)” introduces the proposed meta-heuristic algorithm, describing its working principles and how it is applied to address the scheduling problem. In Section “[Studied system](#)”, the case study using the ETAP is discussed, followed by an analysis of the results and performance evaluation of the proposed method. Section “[Simulation and result analysis](#)” offers a comparative analysis with recent works, highlighting the improvements and advantages of the proposed approach. Finally, Section “[Conclusion and future research directions](#)” concludes the paper, summarizing the key findings and suggesting future research directions.

Modeling of microgrid optimal scheduling

The proposed microgrid system model, which is integrated with the main power grid, is depicted in Fig. 1. This model illustrates how the microgrid operates alongside the main utility grid to enhance power supply reliability and manage diverse load conditions. The microgrid consists of various distributed generation (DG) sources, including solar photovoltaic (PV) systems, wind farms, and energy storage systems (ESS). These DG units are strategically placed on different buses within the network to optimize power generation and distribution³⁹. The incorporation of multiple DGs not only facilitates a more resilient energy supply but also enables the integration of renewable energy sources, contributing to the overall sustainability of the power system.

In the microgrid, load demands are classified into three categories: non-curtable, deferrable, and curtable loads⁴⁰. Non-curtable loads represent critical energy demands that cannot be adjusted or interrupted under normal operating conditions. Examples of these loads include essential services, such as healthcare facilities, data centers, and other critical infrastructure, which require a constant power supply to function effectively.

Deferrable and curtable loads, on the other hand, offer more flexibility in their management. Deferrable loads can be adjusted or shifted in time based on their priority and the available energy supply⁴¹. For instance, charging of electric vehicles or operation of heating, ventilation, and air conditioning (HVAC) systems can be scheduled during periods of low demand or lower electricity tariffs. Curtable loads can be reduced or disconnected temporarily, depending on the importance of the loads and varying electricity tariffs. This flexibility in load management allows the microgrid to optimize energy use and reduce costs, particularly during peak demand periods or when renewable generation fluctuates due to weather conditions⁴². This categorization of loads is crucial for developing effective power scheduling strategies that meet both technical constraints and economic objectives in the microgrid environment⁴³.

In a microgrid, generation sources are classified into two categories: controllable and non-controllable. Controllable sources are those that the microgrid technical controller can actively manage to ensure stable and optimal operation. These sources include dispatchable generators such as diesel generators or energy storage systems (ESS) like batteries. The output power of these sources can be adjusted in real-time to match the power demand, compensate for fluctuations in renewable energy generation, and meet various operational constraints, such as voltage regulation and frequency stability.

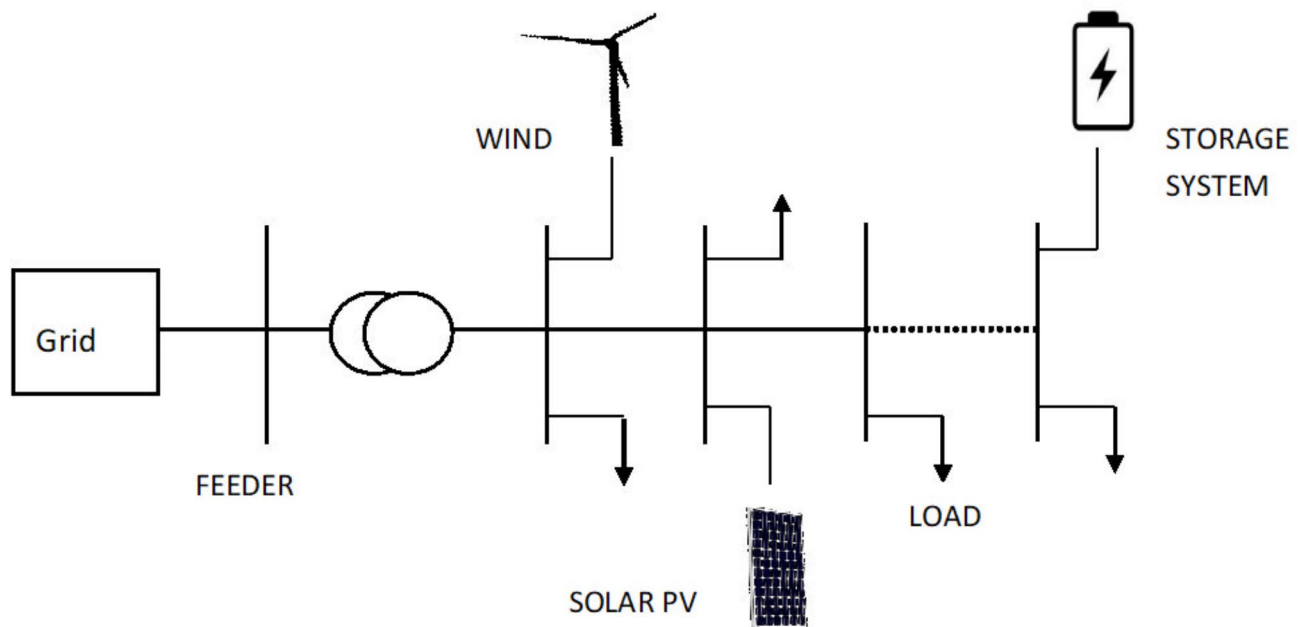


Fig. 1. A single line diagram of microgrid.

In contrast, non-controllable sources are those whose output cannot be directly controlled by the microgrid's technical controller, as their input energy is variable and often unpredictable. These sources are primarily renewable energy resources, such as solar photovoltaic (PV) systems and wind turbines, which generate power based on weather conditions like sunlight and wind speed. Since these sources provide a changeable output, their integration into the microgrid introduces complexities in maintaining a balanced and reliable power supply. Despite their non-dispatchable nature, these DG units can still be utilized strategically in power management. For instance, they can support load shifting based on the scheduling of other controllable generation sources or storage systems to align power supply with demand more effectively.

The load scheduling model of the proposed microgrid is outlined in Fig. 2. This model addresses the power scheduling problem by dividing it into a main problem and several sub-problems. The main problem corresponds to the grid-connected mode, where the microgrid operates in coordination with the main utility grid to supply power to its loads. In this mode, the microgrid can draw power from the grid or inject surplus power into it, depending on generation and consumption conditions.

The sub-problems, on the other hand, represent scenarios where the microgrid operates independently of the main grid. These include islanded mode, where the microgrid must solely rely on its internal DGs and storage systems to meet its load demands. Another sub-problem scenario is grid-connected mode with one DG outage, which simulates the failure of a generation unit, such as a solar plant, due to unfavorable weather conditions or technical faults. This classification into the main and sub-problems helps in designing a robust load scheduling strategy that adapts to various operating conditions, ensuring a reliable power supply while optimizing costs and resource utilization.

The main problem in the proposed model involves calculating the power commitment and dispatch of the available controllable sources within the microgrid. This includes deciding how much power each controllable source, such as diesel generators or battery storage, should produce at any given time to meet the load demands efficiently. Additionally, it involves scheduling various types of loads non-curtable, deferrable, and curtable to align with both the generation capabilities and cost-effectiveness. Another critical aspect of the main problem is managing the active power exchange between the microgrid and the main grid. This exchange can include drawing power from the main grid during shortages within the microgrid or feeding excess power back to the grid when there is a surplus, thus optimizing overall system performance and economic benefits.

The sub-problem, on the other hand, focuses on the microgrid's operation in scenarios where it must rely solely on its internal DG units and ESS. Specifically, this involves calculating an optimal power schedule to ensure that the DGs are sufficient to meet the load requirements when the microgrid is operating in an islanded mode or under conditions of DG outages. The sub-problem evaluates the microgrid's ability to continuously supply power to its loads, considering factors such as fluctuating renewable generation and varying load demands. This ensures that even in isolated conditions, the microgrid can maintain a balanced power supply and address any potential issues, such as voltage instability or power shortages.

The overarching goal of optimal power scheduling in both the main problem and sub-problem is to meet the total load demand at the lowest possible cost, without violating the operational constraints of the system. These constraints include maintaining voltage levels within permissible limits, respecting power flow capacities of transmission lines, and managing the dynamic nature of renewable energy sources. By optimizing the dispatch

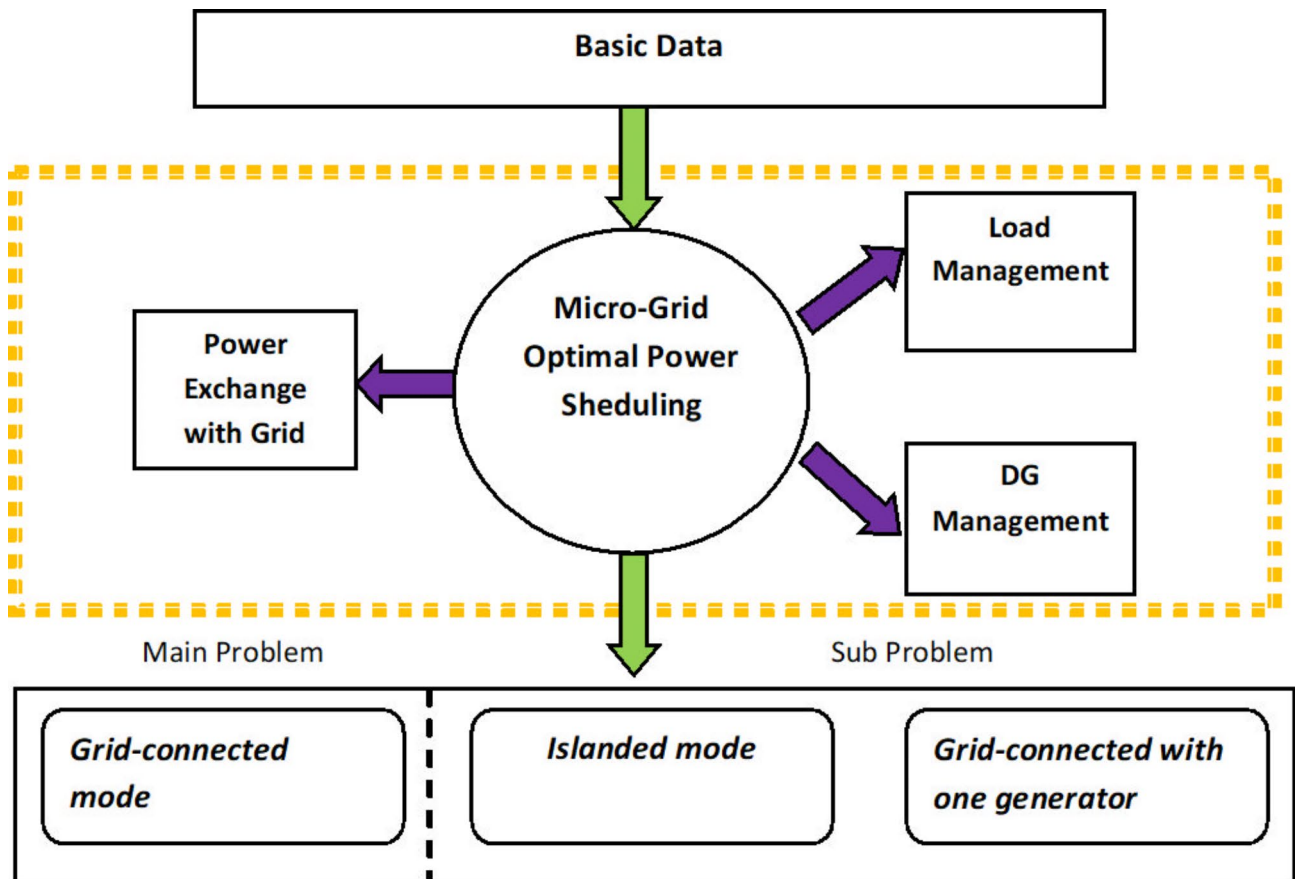


Fig. 2. Flow chart for proposed microgrid power scheduling model.

of controllable sources and scheduling of different loads, the proposed approach aims to enhance the efficiency, reliability, and economic performance of the microgrid under various operating conditions.

Problem formulation

The objective of the microgrid operation is to improve the voltage regulation (VR) and minimize the total operating cost (TOC) of microgrid is formulated as follows.

$$F(x) = \text{Max} \sum_t \frac{(V_S - V_R)}{V_S} + \text{Min} \sum_{i=1}^T (K \times P_{i, \text{Loss}} + K \times \int_0^t P dt) \quad (1)$$

The first term in the objective is the voltage regulation of microgrid, which includes sending end voltage and receiving end voltage. The second term is the total operating cost of the microgrid. Where K - the fixed cost per kW, K = 60 rupees fixed by Tamil Nadu electricity regulatory commission (TNERC)⁴⁴ based on low tension tariff I-C, II-A, II-B (1), II-B (2), II-C. The operating cost of the MG system is based on the following criterion: The energy is described as Eq. (2),

$$\int_0^t P dt = E \quad (2)$$

where the energy consumption during light load is denoted as E*4 (kWh), the energy consumption during normal load is denoted as E*8 (kWh), the energy consumption during peak load is indicated as E*12 (kWh). The operating cost per month is calculated by E (kWh)* K. The objective is subject to equality and inequality constraints, as follows:

The equality constraint is given by the following equation:

$$P_{\text{Swing}} = \sum_{i=0}^n P_{i, \text{Loss}} + \sum_{K=0}^n P_d(k) \quad (3)$$

where P_{Swing} is the active power swing bus n , $P_{i, Loss}$ is the real power loss at bus i and P_d is the demand of the active power at bus n . The inequality constraint is given by the following equation. The magnitude of each bus voltage must be limited by the following equation⁴⁵:

$$V^{\min} \leq V \leq V^{\max} \quad (4)$$

A. Network modeling

The MG network is modeled by the following recursive equations⁴⁶. The real power, reactive power, and voltage of $i+1$ bus can be described by the Eqs. (5), (6), and (7).

$$P_{i+1} = P_i - r_i \frac{(P_i^2 + Q_i^2)}{V_i^2} - P_{L(i+1)} \quad (5)$$

$$Q_{i+1} = Q_i - x_i \frac{(P_i^2 + Q_i^2)}{V_i^2} - Q_{L(i+1)} \quad (6)$$

$$V_{i+1}^2 = V_i^2 - 2(r_i P_i + x_i Q_i) + r_i \frac{(P_i^2 + Q_i^2)(r_i^2 + x_i^2)}{V_i^2} \quad (7)$$

The above set of recursive equations is used to model the power flow of the MG system. Where P_i and Q_i are the real and reactive line power of i^{th} bus respectively; $P_{L(i+1)}$ and $Q_{L(i+1)}$ are the real and reactive load powers of the $i+1$ th bus. The resistance and reactance of the line section of the bus i are denoted by r_i and x_i . The real power loss of P_{i+1} , Loss may be computed using Eq. (8)

$$P_{i+1, Loss} = r_i \frac{(P_i^2 + Q_i^2)}{V_i^2} \quad (8)$$

Crow search algorithm

The new meta-heuristic approach, known as the Crow Search Algorithm (CSA), was proposed by Alireza in 2016⁴⁷. This innovative algorithm is inspired by the natural behavior of crows, which are known for their remarkable intelligence. Crows have the ability to hide their excess food in various locations and then retrieve it when needed. This food-hiding behavior forms the basis of the algorithm's design.

Crows are considered one of the most intelligent bird species, possessing the largest brain-to-body size ratio among birds. They exhibit advanced cognitive abilities, including the capacity to remember other crows' faces, recognize when they are being observed, and adjust their behavior accordingly to avoid losing their hidden food. Moreover, they are known to hide food in multiple places, using complex strategies to protect their resources from potential thieves. Crows can even mislead other birds by pretending to hide food in one location while secretly storing it elsewhere.

In the context of the CSA, this food-hiding and retrieval behavior is modeled to explore and exploit the search space in optimization problems effectively. The algorithm simulates crows' actions of storing and recalling information (similar to how crows remember their food hiding spots) to find optimal solutions. The crow's intelligence in keeping track of locations and avoiding detection is analogized to the algorithm's ability to explore different potential solutions and adapt to changes in the problem space. By drawing inspiration from this natural behavior, the CSA is designed to balance both local and global search strategies, enhancing its ability to solve complex optimization problems like power scheduling in microgrids. Crow search algorithm works based on the below four rules.

1. Crows live in cluster;
2. Crows recall the position of the concealed place;
3. Crows can join each other while thieving food;
4. Crows keep the opportunity of grabbing the food.

Assuming an N number of crows of the cluster, the i^{th} number of a crow at iteration k is x_i^k . Each crow ' i ' follows the hiding place of the food is memorized. The hiding place of the food followed by crow i is memorized. Each crow has a memory in which the position of its hiding place is memorized is given as m_j^k . Assume at the time of the iteration crow ' j ' want to visit the food hiding place of crow ' i ', which is denoted by m_j^k . At a similar time, crow ' i ' decides to follow crow ' j '. In this regard, two cases may happen.

Case1: The crow thief position is updated by

$$x_i^{k+1} = x_i^k + r_i * fl_i^k * (m_j^k - x_i^k) \quad (9)$$

where r_i is the random number between 0 and 1. fl_i^k is the flight length of crow ' i ' at iteration k .

Case2: If crow ' j ' realized that the crow ' i ' is following, crow ' j ' has to give the wrong impression about crow ' i ' such that it will not find its food position. Then, crow ' i ' has to move again to find another crow to follow and so shift to a new random position. This is the best location that crow ' i ' has obtained so far.

In CSA, the scenario is calculated by the following expression:

$$\begin{aligned}
 & ifl_j \geq P_j^k \quad \text{Update position by eqn. 9} \\
 & \text{else} \\
 & \quad \text{update to random position}
 \end{aligned} \tag{10}$$

where r is a random number between 0 and 1, and P_j^k is the probability of awareness of crow ' j ' at iteration k . The value of ' fl ' is participating important role in the search process. Here smaller values of ' fl ' lead to local search and larger values lead to a global search. Indeed, we have to make a balance between local and global searching of the search space. The ' fl ' value is also helping to achieve convergences of the search process. The following step by step procedure is used for the performance of CSA.

Step 1: Initialize the problem and variable parameters

The variable parameters of CSA are size of the flock (N), the maximum number of iterations (i^{max}), flight length (fl), and the probability of awareness (P_j^k) are respected³⁴.

Step 2: Initialize the position and crow memory

First, initialize each crow's memory. While at the initial position, the crows have no incidents, it is imagined that they have secreted their foods at their initial conditions. N numbers of crows are randomly placed in a matrix search space as the flock memory. The memory of crow is denoted as,

$$\text{Memory} = \begin{bmatrix} m_1^1 & m_2^1 & \dots & m_n^1 \\ m_1^2 & m_2^2 & \dots & m_n^2 \\ \dots & \dots & \dots & \dots \\ m_1^n & m_2^n & \dots & m_n^n \end{bmatrix} \tag{11}$$

Step 3: Estimate the objective function

The excellence of each crow and its location is calculated by adding the decision variables into the objective.

Step 4: Produce crow new position

Produce the crow's new position in the search space is based on Eq. (9). This procedure is frequent for all the crows.

Step 5: Check the possibility of new positions

The possibility of the new place of every crow is verified. If the new place of a crow is possible, the crow updates its place. Otherwise, the crow continues in the current place and does not shift to the produced new place.

Step 6: Estimate the objective function of new position.

The objective function value for the new place of each crow is calculated.

Step 7: Update the crow memory

It is cleared that if the objective function value of the new place of a crow is better than the objective function value of the memory place, the crow replaced its memory by the new place.

Step 8: Check termination condition

Step 4 to step 7 is repeated until i^{max} is achieved. When the termination condition is met, the best place of the memory in terms of the fitness function value is accounted as the output of the optimal solving problem. The algorithm is applied to the proposed methodology is shown in Fig. 3.

Studied system

The case study for this research is conducted on a 31-bus system operated by Tamil Nadu Generation and Distribution Corporation Limited (TANGEDCO) in the Kattur feeder of the Ambikapuram substation, located in Tiruchirappalli district, Tamil Nadu, India. The system operates at an 11 kV voltage level, with a total capacity of 16 MVA and a frequency of 50 Hz. The network experiences a real power demand of 9790 kW and a reactive power demand of 4744 kVar, making it a robust system that requires careful power management. The detailed line and load data for this system are provided in Appendix A (Supplementary File). In the case study, the penalty factor for decreasing the adjustable load is determined to be 25.

This distribution network supplies critical industrial and healthcare loads, which necessitate a continuous and reliable power supply along with stringent voltage regulation to maintain their operational stability. However, the system currently faces significant challenges, as it exhibits poor voltage regulation of 8.452%, coupled with a high real power loss. Additionally, the network has a diversity factor of 1.5, indicating the variation in load patterns across different times of the day and further complicating the task of power scheduling and voltage control.

To address these issues, the proposed microgrid system is modeled and analyzed using the Electrical Transient Analyzer Program (ETAP) software. The ETAP environment allows for the detailed simulation and analysis of the microgrid's operation under different conditions. Also, enabling an assessment of various power scheduling strategies, including the incorporation of distributed generation (DG) sources and the proposed crow search algorithm. Through this modeling, the case study aims to demonstrate how the optimal scheduling of power in a microgrid can improve voltage regulation and reduce real power losses, thereby enhancing the overall reliability and efficiency of the distribution system.

The three cases of operations are illustrated to analyze the performance of the microgrid system.

Case 1: grid-connected mode.

Case 2: islanded mode.

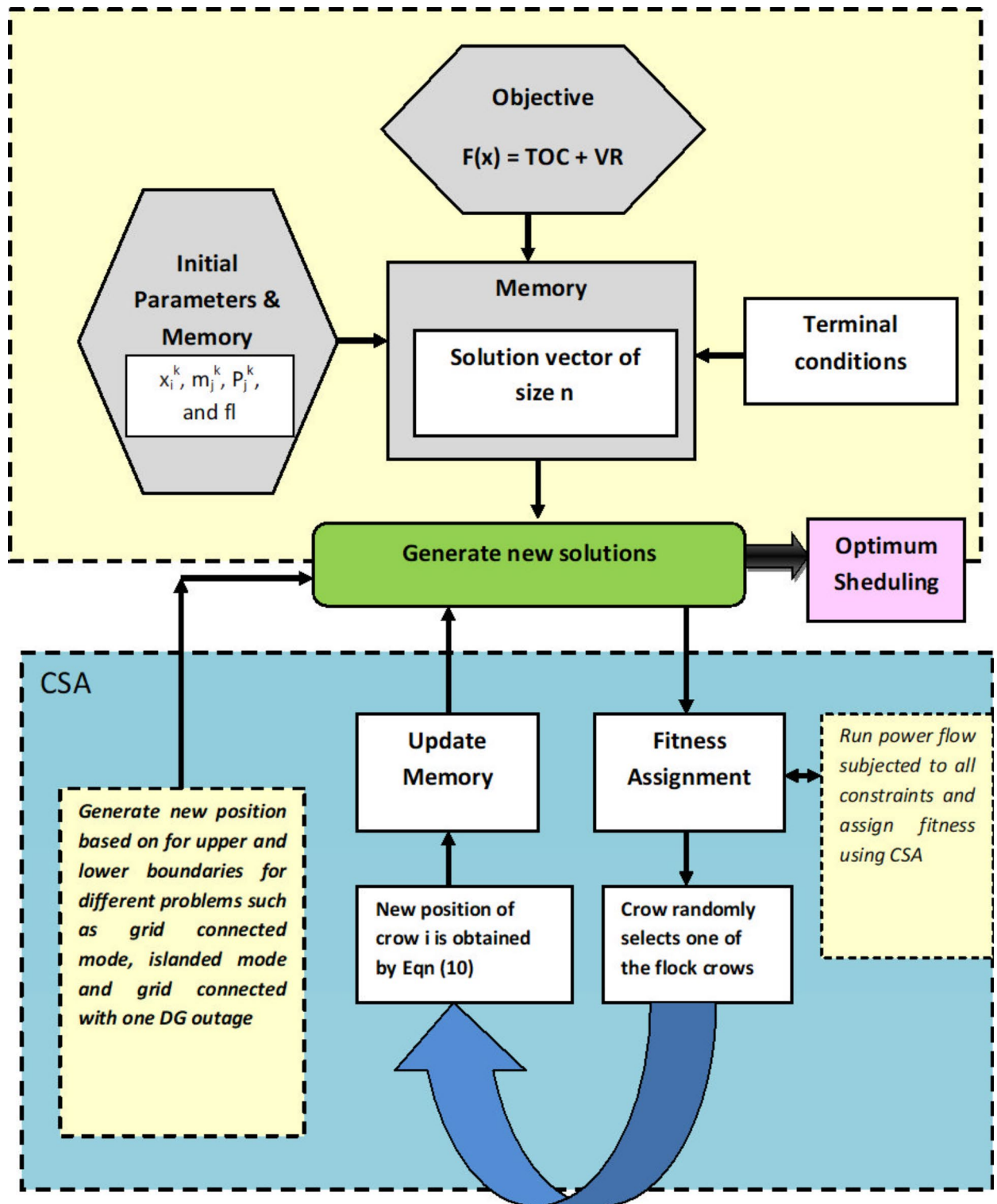


Fig. 3. Flowchart for proposed algorithm is applied to power scheduling problem.

Case 3: grid-connected with one generator outage.

Case 1: Grid-connected mode:

In the grid-connected mode, the microgrid operates in conjunction with the main utility grid to meet the overall power demand. For this mode to function effectively, the following conditions must be satisfied: (i) the main grid must supply the total real power demand required by all loads within the microgrid; (ii) the frequencies of the main grid and the distributed DGs within the microgrid must be synchronized to maintain

system stability. Synchronization is crucial to ensure a seamless power exchange between the grid and the microgrid, avoiding issues like power quality degradation or unintentional islanding.

In this mode, both the main grid and the microgrid's DGs, such as solar photovoltaic (PV) systems, wind turbines, and other controllable sources, work collaboratively to meet the system's load demand. The grid acts as the primary source of power, with DGs supplementing the supply, thus optimizing resource usage. A static transfer switch (STS) is employed to facilitate the connection between the microgrid and the main utility grid. The STS is normally in a closed position, allowing power to flow freely between the grid and the microgrid, depending on demand conditions and generation availability within the microgrid.

Operating in the grid-connected mode offers several benefits, including improved voltage regulation and the ability to draw power from the main grid during periods when internal DG output is insufficient. This configuration also allows for surplus power generated within the microgrid to be fed back into the main grid, potentially lowering operational costs and enhancing the efficiency of the overall power system.

For TANGEDCO system, Power interchange in grid-connected mode decides nature of a microgrid (as a source or as a load) from perspective of a TANGEDCO system. Presence of DG in a microgrid introduces flexibility in microgrid operation. Proper coordination between uses of available DG controls the load demand–supply gap in a microgrid making it more advantageous. A microgrid can absorb the surplus power in the grid during off-peak hours in the grid-connected mode. Penetration of power by a microgrid in a TANGEDCO system is assessed as penetration ratio. Penetration ratio decides the power fed by a microgrid in TANGEDCO system. Penetration Ratio of a Microgrid is denoted by.

$$\lambda = \frac{P_{\mu G}^{in}}{P_D}$$

Where $P_{\mu G}^{in}$ is the total power exported by a microgrid in TANGEDCO system at Point of Common Coupling (PCC), P_D is the total active power demand TANGEDCO system including microgrid demand. In the case study, this mode is examined to assess how effectively the grid and DGs can coordinate to maintain a stable and reliable power supply under varying load conditions.

Case 2: Islanded mode:

In the absence of a connection to the utility grid, the microgrid operates in islanded mode, relying solely on its internal power generation resources to meet the load demand. In this mode, the STS is opened, effectively isolating the microgrid from the main grid, which may occur during grid failures or maintenance activities. A sustained islanded mode operation requires that the sum of the power outputs from all distributed DGs within the microgrid meets the critical real power demand of the loads. During islanded operation, each DG in the microgrid is capable of generating up to 1500 kW. However, in this case study, the total generation capacity of the DGs is less than the total load demand of the microgrid, creating a challenging scenario. This shortfall requires careful load management and optimal power scheduling to prioritize critical loads and ensure a continuous supply. Non-essential or deferrable loads may need to be adjusted or curtailed to maintain the balance between generation and demand, preventing power shortages and potential voltage instability. This mode of operation emphasizes the importance of advanced scheduling algorithms and effective use of ESS to maximize the use of available DG capacity. The main objective in islanded mode is to ensure that the microgrid can operate autonomously, providing a reliable power supply to critical loads, even when generation resources are constrained. During this mode, the penetration ratio is zero. The case study explores how the proposed approach manages power distribution, schedules loads, and maintains system stability under these islanded conditions.

Case 3: Grid-connected with one generator outage:

In this operating scenario, the microgrid functions similarly to the standard grid-connected mode; however, one of the distributed generation units (specifically, the solar plant—DG1) fails due to adverse weather conditions. This failure simulates a common real-world situation where renewable energy sources, such as solar power, become unavailable due to factors like cloudy weather or storms. Despite the loss of DG1, the microgrid must still meet the total load demand. To compensate for the shortfall, the remaining distributed generation units (DGs) within the microgrid, along with the utility grid, work together to supply the necessary power. In this case, the microgrid controller dynamically adjusts the power output of the remaining DGs and manages power imports from the main grid to ensure a continuous and stable power supply. This scenario emphasizes the need for flexible and responsive power scheduling, as the microgrid must adapt to the sudden loss of a generation source. The ability to effectively handle such outages while maintaining voltage regulation and minimizing operational costs is crucial for reliable microgrid operation. The case study explores how the proposed scheduling algorithm coordinates with the utility grid and the remaining DGs to satisfy the load demand under these conditions, highlighting the robustness and adaptability of the microgrid in the face of generation uncertainties.

Simulation and result analysis

The TANGEDCO 31-bus test system was developed to reflect actual operational conditions, taking into account load forecasts for a specific day. This setup was crucial for evaluating the effectiveness of the proposed methodology under realistic scenarios. The test system encompasses a variety of loads and distributed generation (DG) units, creating a diverse environment for analyzing the potential of the microgrid's optimal power scheduling strategy. In the islanded mode, the optimal power scheduling was executed using the proposed Crow Search Algorithm (CSA). To thoroughly assess the performance of the microgrid under varying conditions, the distributed generation sources (DGs) were categorized into three different levels of generation: full generation (FG), half generation (HG), and one-fourth generation (OFG). This categorization simulates different scenarios that the microgrid might encounter in real-world operation, such as varying levels of solar or wind energy production due to weather changes. Additionally, the load conditions within the microgrid were classified into three categories: light load, normal load, and peak load. These classifications represent different times of the

day or operational periods, providing a comprehensive analysis of how the microgrid performs under varying demand levels. For the case study, the grid voltage was set at 1.05 per unit (p.u) to enhance voltage regulation and ensure stable power supply throughout the network. Figure 4 illustrates the optimal structure of this real-time microgrid system, showcasing how the different DGs and load types interact under the specified conditions. By using these varying operational scenarios, the study effectively demonstrates the capability of the proposed methodology to adapt to diverse and dynamic environments, ensuring both cost-efficient and reliable microgrid operation.

In the islanded condition, the optimal load scheduling using the proposed CSA was applied and demonstrated for a specific day, as detailed in Table 2. On this particular day, the microgrid's total load demand amounted to 8534 kW. To meet this demand, both conventional power generation and DG units were considered. The conventional power generators were capable of producing 9790 kW, while the combined DGs (DG1, DG2, and DG3) had an optimal generation capacity of 5500 kW. However, in the islanded mode, the total available DG capacity is constrained to 4500 kW, which is less than the overall load demand.

Despite this limitation, the proposed CSA was employed to efficiently schedule the power supply within the proposed microgrid. Through strategic load management and prioritization, the algorithm ensured that the essential load demands were met, even with the reduced generation capacity. The CSA optimized the distribution of the available power among different load types, thereby maintaining system stability and reliability.

Table 2 provides a detailed overview of the optimal power allocation to 12 selected load demands out of the 31 total loads in the system. The selected loads include seven non-curtailable loads, which are critical and must be supplied continuously, four deferrable loads that can be adjusted based on the importance and available generation, and one curtailable load that can be temporarily disconnected if necessary. By optimizing the scheduling of these loads, the microgrid could operate effectively within the islanded condition, balancing the reduced generation capacity with the necessary power demands. This demonstrates the capability of the CSA to manage and prioritize load scheduling efficiently, ensuring that vital operations are not interrupted during periods of limited power generation.

In the proposed method, minimizing the total operating cost (TOC) serves as the primary objective function. This objective ensures that the microgrid operates economically while meeting the necessary load demands and maintaining system reliability. Table 3 presents a detailed analysis of the total power loss, associated loss costs, and the TOC for a typical day across three different operational scenarios: grid-connected mode (Case 1), islanded mode (Case 2), and grid-connected mode with one generator outage (Case 3).

From the observations in Table 3, it is evident that the Case 2 achieves the lowest TOC compared to the other two modes. Accordingly, under light load with full generation condition the case 2 results compared with

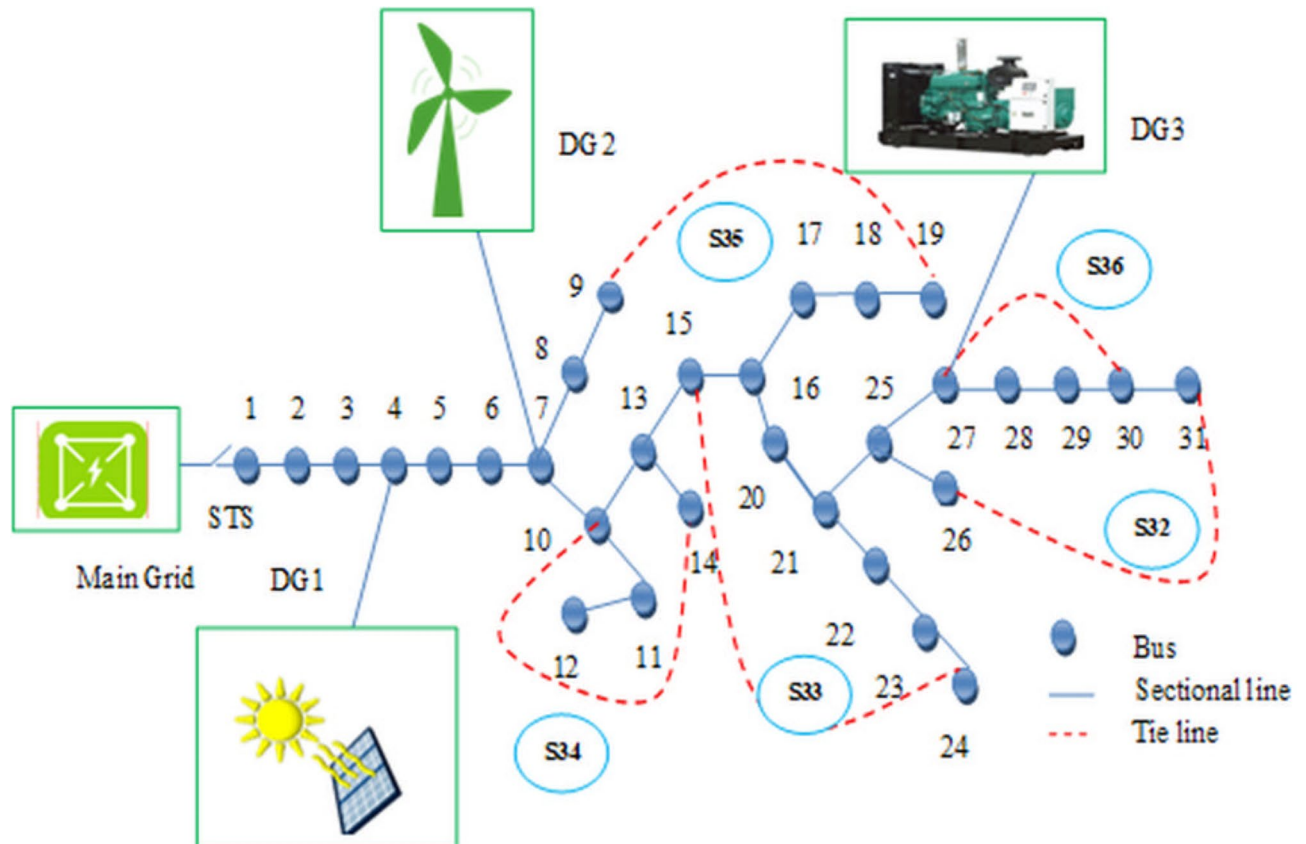


Fig. 4. Optimal structure of practical system with DG.

Islanded mode % of loading									
Loads	Loading configuration								
	Light load			Normal load			Peak load		
	FG	HG	OFG	FG	HG	OFG	FG	HG	OFG
Non curtailable loads									
Vivekanandha Nagar 1	49.4	49.7	50.1	69.2	69.8	70.3	99.3	100	100.8
Kattur III	49.4	49.7	50	69.3	69.7	70.1	99.3	99.9	100.5
Perriyar Nagar	49.4	49.7	50	69.3	69.7	70.1	99.3	99.9	100.5
Gandhi Nagar 1	49.5	49.7	50	69.4	69.8	70.1	99.3	99.9	100.5
Balaji Nagar 6	50	50	50.1	70	70.1	70.2	100.2	100.4	100.5
Balaji Nagar 5	50	50	50.1	70	70.1	70.2	100.2	100.4	100.5
Balaji Nagar 7	50	50	50.1	70	70.1	70.3	100.2	100.4	100.6
Differable loads									
Kattur VI	49.3	49.7	50	69.2	69.7	70.2	99.2	99.9	100.6
Kattur IV	49.5	49.7	50	69.4	69.7	70.1	99.3	99.9	100.5
Kailash Nagar 4	49.7	49.8	50	69.6	69.9	70.1	99.7	100	100.4
Balaji Nagar 4	49.9	50	50.1	69.9	70	70.1	100	100.2	100.3
Curtailable loads									
Kailash Nagar 2	50	50	50	70	70	70	90	100	100

Table 2. Optimal load scheduling.

Loading condition	Generation type	Grid connected mode (case 1)			Islanded mode (case 2)			One DG outage (DG1) (case 3)		
		Total loss (kW)	Loss cost (Rs)	TOC (Rs)	Total loss (kW)	Loss cost (Rs)	TOC (Rs)	Total loss (kW)	Loss cost (Rs)	TOC (Rs)
Light load	FG	24	1440	341,840	16	960	268,080	30	1800	463,800
	HG	85	5100	682,140	5	300	262,380	104	6240	846,000
	OFG	239	14,340	1,246,500	4	240	262,080	239	14,340	1,246,500
Normal load	FG	93	5580	1,263,660	18	1080	740,760	132	7920	1,920,720
	HG	245	14,700	2,384,940	3	180	732,180	278	16,680	2,719,560
	OFG	495	29,700	3,556,740	6	360	734,220	495	29,700	3,556,740
Peak load	FG	373	22,380	4,226,740	13	780	1,580,280	457	27,420	5,245,260
	HG	679	40,740	6,023,220	94	5640	1,573,080	738	44,280	6,544,440
	OFG	1115	66,900	7,918,500	161	9660	1,582,140	1115	66,900	7,918,500

Table 3. Minimization of TOC.

case 1 and case 3, the TOC is significantly reduced by 21.5% and 42.1%. Similarly, under normal load with full generation condition the case 2 results compared with case 1 and case 3, the TOC is significantly reduced by 22.2% and 61.4%. Also, under peak load with full generation condition, the case 2 results compared with case 1 and case 3, the TOC is significantly reduced by 62.6% and 69.8%.

This reduction in TOC highlights the effectiveness of the proposed approach in optimizing power distribution and load scheduling within the microgrid when it operates independently from the main grid. By prioritizing the dispatch of DG units and adjusting load demand, the CSA minimizes unnecessary power losses and operating costs in the islanded condition.

Furthermore, the table indicates significant reductions in power losses under the Full Generation (FG) condition. Specifically, when comparing the Case 2 with the Case 1, the loss reductions are 66.54%, 62.12%, and 69.12% at light, normal, and peak load levels, respectively. Similarly, when comparing the Case 3 to Case 1, the loss reductions are observed to be 81.12%, 76.98%, and 82.83% at the same load levels. These figures clearly demonstrate that the proposed method not only minimizes the operating costs but also substantially reduces power losses across various operational scenarios.

This performance enhancement is particularly significant because reducing power losses directly contributes to lower operational costs and improved voltage regulation, ultimately leading to more efficient and reliable microgrid operations. The ability of the proposed method to maintain such efficiency under different conditions showcases its robustness and adaptability, validating its practical application in real-time microgrid management.

Figures 5a–c and 6a–c illustrate the real power loss at each branch and the voltage profile at each bus in the case study system under light load conditions. Similarly, Figs. 7a–c and 8a–c present the real power loss and voltage profiles under normal load conditions. These figures provide a detailed visual comparison of the power losses and voltage levels across the microgrid for the three operational scenarios: Case 1, Case 2, and Case 3.

From these observations, it is evident that the real power loss at each branch is minimized in the Case 2 compared to the other two modes. This reduction in power loss can be attributed to the optimized load scheduling

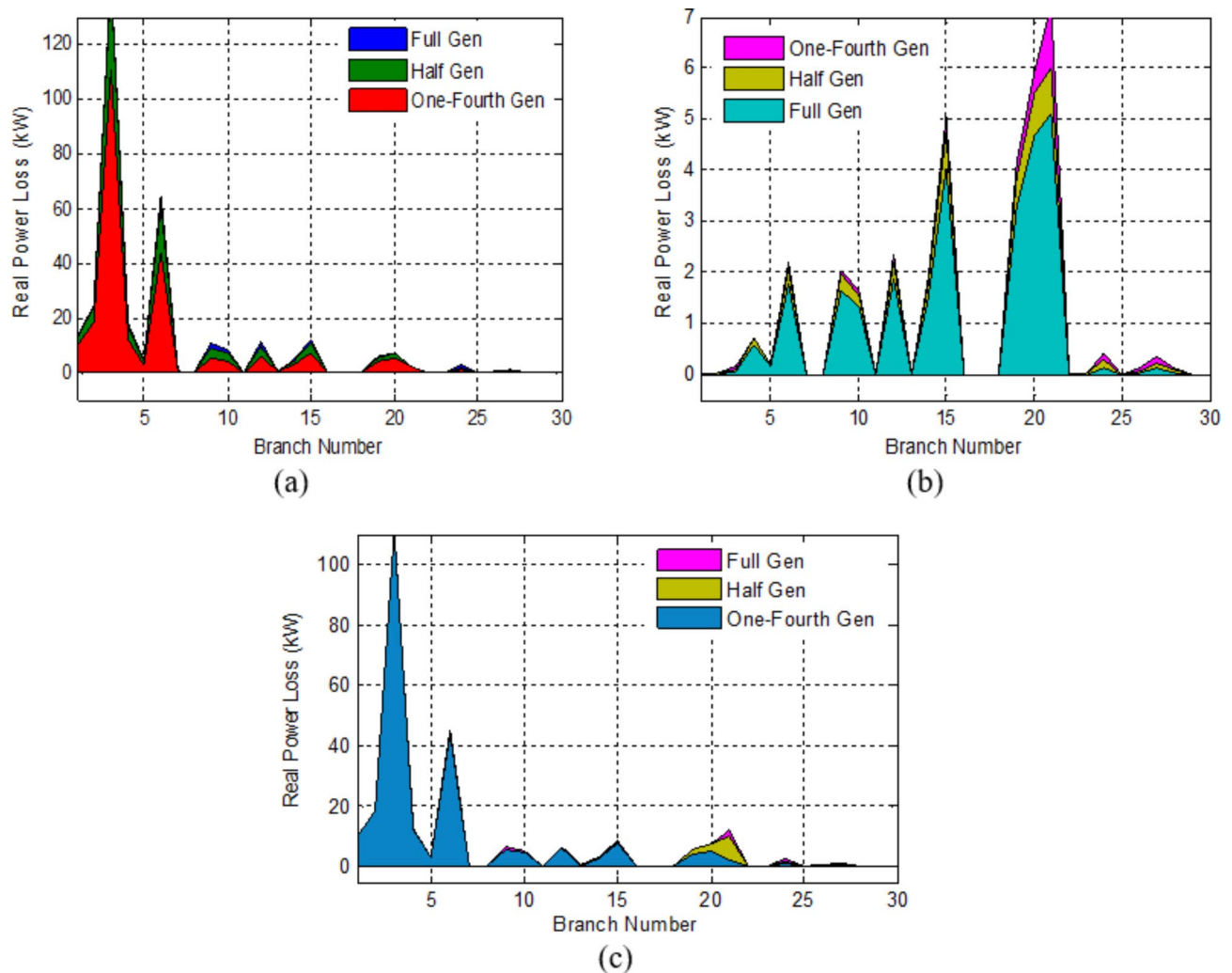


Fig. 5. Real power loss at each branch under light load (a) case 1 (b) case 2 (c) case 3.

and power distribution strategy achieved through the proposed CSA. By efficiently dispatching the available DG units and managing the loads within the microgrid, the CSA minimizes unnecessary power dissipation, thereby enhancing the system's overall efficiency.

Additionally, the voltage profile depicted in the figures indicates improved voltage regulation in Case 2, which is crucial for maintaining the stability and reliability of the microgrid, especially when operating independently from the main grid. This improvement underscores the effectiveness of the proposed scheduling method in managing voltage levels across different buses, ensuring they remain within permissible limits.

However, these figures also reveal that if the DG capacity in the microgrid is low, the real power loss increases. This is particularly noticeable in Case 3, where one of the DG units is out of service. In such situations, the microgrid relies heavily on the remaining DGs and, potentially, on power drawn from the main grid, resulting in increased power losses. Therefore, maintaining an adequate DG capacity is essential for minimizing losses and enhancing the microgrid's performance. The ability of the proposed method to adapt to varying DG capacities and load conditions demonstrates its robustness in optimizing microgrid operations under different scenarios.

In each operational scenario, except for Case 2, a reduction in distributed generation (DG) capacity from Full Generation (FG) to One-Fourth Generation (OFG) resulted in a simultaneous increase in both total real power loss and total operating cost (TOC). This increase is due to the limited availability of generation resources, which forces the microgrid to rely more on less efficient power scheduling and load management strategies. Consequently, power losses rise as the system attempts to balance supply and demand with reduced generation capacity, leading to a higher TOC.

The TOC was calculated based on the energy consumption at each bus within the microgrid, considering three distinct load levels: light, normal, and peak. In this study, the TOC reached its maximum in Case 1, indicating that the grid's reliance on both DGs and external power sources without optimized scheduling results in higher operational expenses. Conversely, the TOC was minimized in Case 2, demonstrating the effectiveness of the CSA in optimizing power scheduling within the microgrid and efficiently utilizing the available DG capacity.

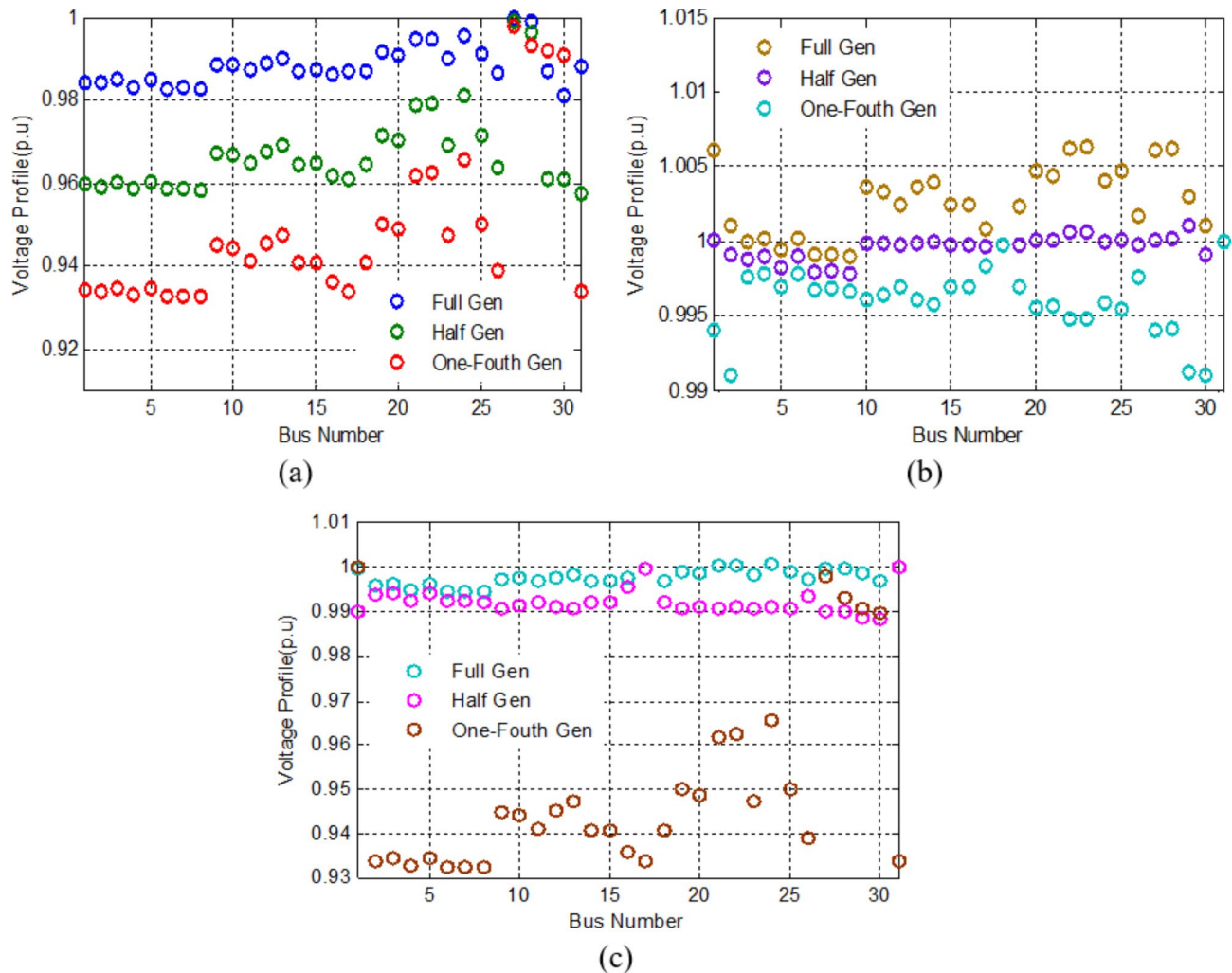


Fig. 6. Voltage profile at each bus under light load (a) case 1 (b) case 2 (c) case 3.

Figure 9a–c depict the TOC for the microgrid system under different load conditions—light, normal, and peak, respectively. These figures clearly illustrate the relationship between DG capacity and operating costs across the various cases. Notably, in Case 2, the CSA efficiently schedules the loads and dispatches power, thereby reducing power losses and operating costs even as DG capacity changes. This highlights the proposed method's robustness in maintaining economic and reliable microgrid operations across varying conditions.

These findings emphasize the importance of maintaining an adequate DG capacity in the microgrid and implementing optimal scheduling strategies to minimize TOC. The results further validate that the proposed method significantly enhances the microgrid's performance by balancing energy consumption and generation, particularly in islanded operation.

Therefore, the second objective function of this study was the maximization of voltage regulation (VR) within the microgrid. Proper voltage regulation is critical for maintaining power quality and ensuring the stable operation of sensitive loads. Table 4 presents the VR values for a typical day across the three operational scenarios: grid-connected mode (Case 1), islanded mode (Case 2), and grid-connected mode with one DG outage (Case 3).

The results in Table 4 reveal that the maximum VR of 0.86% occurs in the Case 1 under the Full Generation (FG) condition, where all DGs are operating at their full capacity. However, in the Case 2, the VR significantly improves, dropping to just 0.1%. This substantial improvement demonstrates the effectiveness of the proposed Crow Search Algorithm (CSA) in optimizing power scheduling to maintain voltage levels within optimal limits, even when the microgrid is operating independently of the main grid.

As observed from Table 4, Case 2 consistently achieves better VR compared to both Case 1 and Case 3 across various load conditions. This outcome underscores the robustness of the proposed method in managing voltage profiles within the microgrid, particularly when it operates in islanded mode. By ensuring that voltage deviations are minimized, the method contributes to a more stable and reliable power supply, which is crucial for both critical and non-critical loads in the microgrid.

Evaluate the effectiveness of the proposed approach, including power loss reduction, cost savings, and improvements in voltage regulation. The efficiency of proposed approach results is shown in Table 5. These

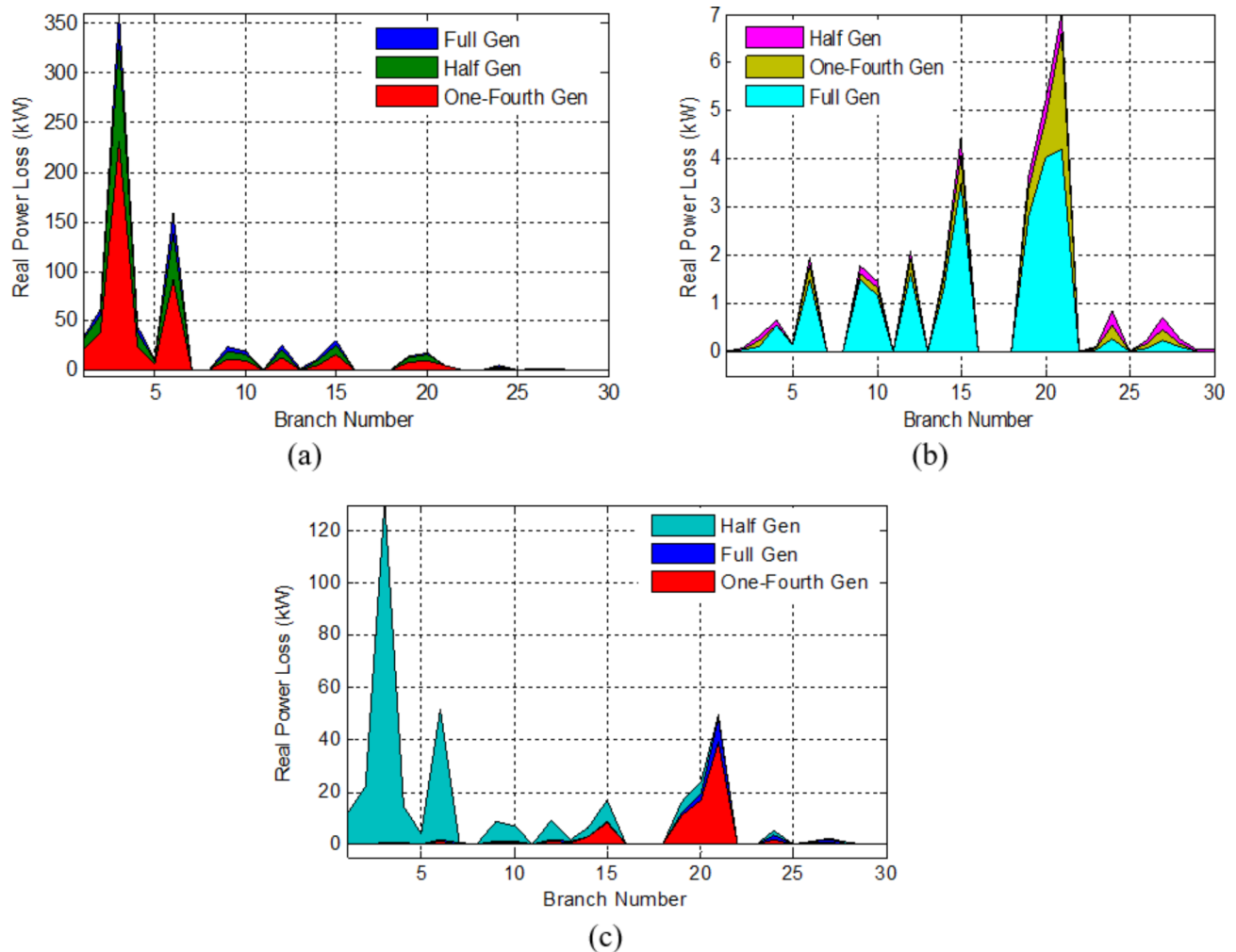


Fig. 7. Real power loss at each branch under normal load (a) case 1 (b) case 2 (c) case 3.

findings validate that the proposed approach not only cost savings but also maximizes voltage regulation, thereby enhancing the overall performance and reliability of the microgrid. The ability to achieve such improvements in voltage regulation, especially in the islanded mode, is essential for microgrids to operate autonomously and sustain power quality under varying conditions.

Compute the corresponding total operating cost, total loss, and voltage regulation of the real time power scheduling under different strategies. For case 1 under normal load condition, the comparison of the proposed technique to with other techniques is shown in Table 6. The results show that proposed CSA have excellent performance. The proposed CSA method has achieved the lowest system loss and operating cost has an advantage of Rs.750,667 than the evolutionary method. In terms of voltage regulation, the proposed CSA method has the maximum, which is 2.6%, which is the greater results compared to the other methods. The proposed approach obtains the best results in terms of total operating cost, total loss, and voltage regulation of the real time power scheduling of all.

Conclusion and future research directions

In this study, an optimal power scheduling method for microgrids was proposed using the Crow Search Algorithm (CSA), with the primary objectives of minimizing the total operating cost (TOC) and maximizing voltage regulation (VR). The methodology was tested on a 31-bus microgrid system in Tamil Nadu, India, under three different operational scenarios: grid-connected mode, islanded mode, and grid-connected mode with one DG outage. The results demonstrated that the proposed method effectively reduced power losses, maintained stable voltage profiles, and minimized TOC, particularly in the islanded mode. The islanded mode (Case 2) proved to be the most efficient in terms of cost and voltage regulation, outperforming both the standard grid-connected mode (Case 1) and the scenario with one DG outage (Case 3). This outcome highlights the robustness and adaptability of the CSA in managing distributed generation (DG) resources and scheduling loads optimally, even under challenging conditions such as generator outages or limited power availability. Furthermore, the results indicated that maintaining an adequate DG capacity is crucial for minimizing power losses and ensuring reliable microgrid operation. The study contributes to the field by incorporating comprehensive modeling of

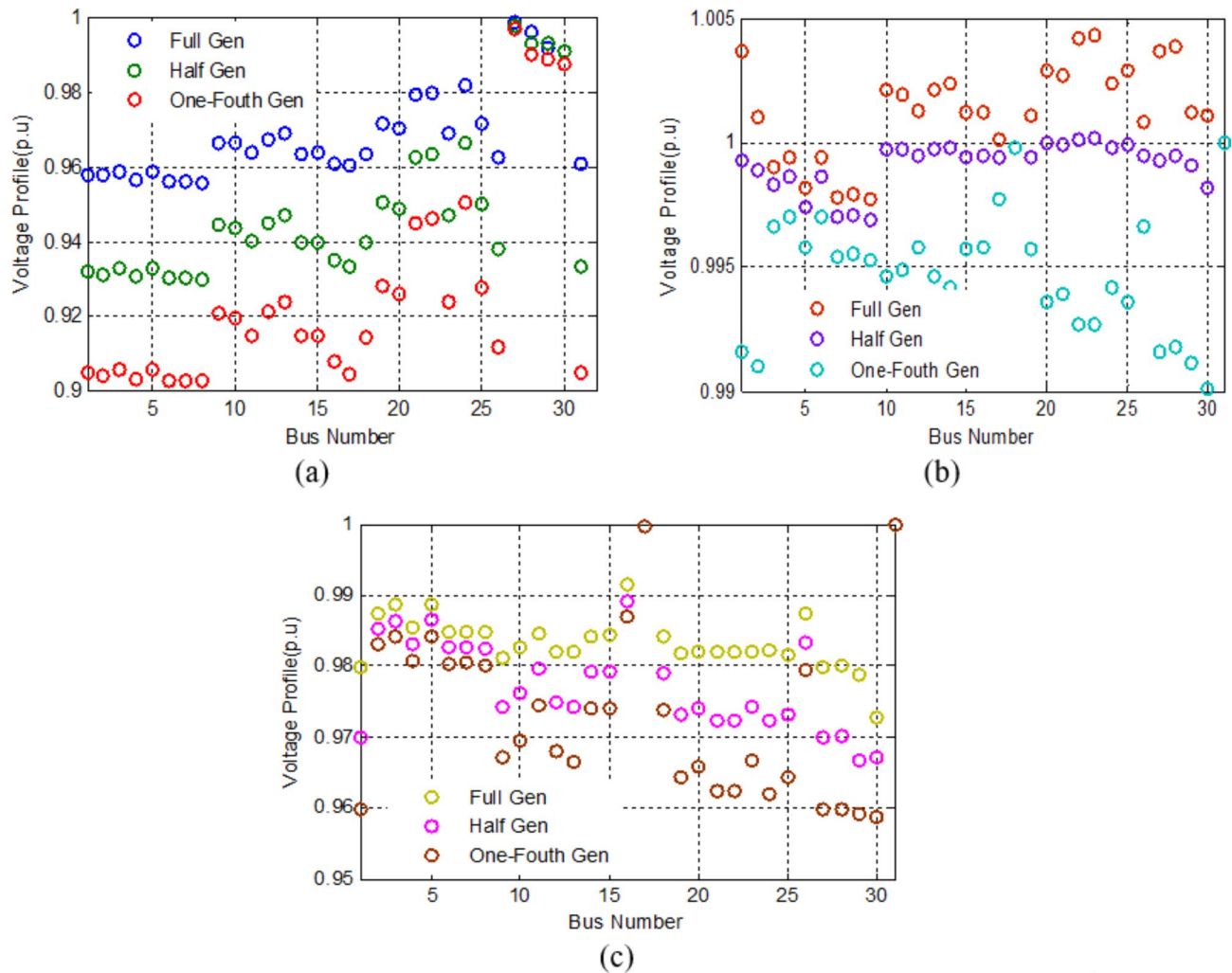


Fig. 8. Voltage profile at each bus under normal load (a) case 1 (b) case 2 (c) case 3.

renewable energy uncertainties, AC optimal power flow (AC-OPF) for a more accurate representation of power dynamics, and effective scheduling strategies using a meta-heuristic approach. Compared with other techniques, the proposed approach increased the cost saving by 44.4% and improvement in voltage regulation by 39.5%. By addressing both TOC minimization and VR maximization, the proposed method enhances the overall performance and reliability of microgrid operations.

While this study has shown that the Crow Search Algorithm (CSA) can effectively optimize power scheduling in microgrids, there are several areas where future research could be helpful. Despite the efficiency of the proposed CSA, this study may also have certain limitations. The proposed approach requires significant historical data, which may be difficult to collect for other large-scale systems. Furthermore, this study only focuses on the economic aspect and voltage regulation of a microgrid power scheduling and assumes to ignore some technical aspects. Future work could use real-time data from smart meters, sensors, and weather forecasts to improve power scheduling. Using up-to-date information can help the microgrid quickly adapt to changes in power demand and renewable energy supply. Future research could include a broader range of uncertainties, such as sudden changes in weather affecting solar or wind power, unexpected load spikes, or equipment failures. This would make the scheduling method more flexible and reliable in real-world situations. Future studies could focus more on energy storage systems (ESS), exploring how advanced storage can support load balancing, improve voltage regulation, and handle the ups and downs of renewable energy.

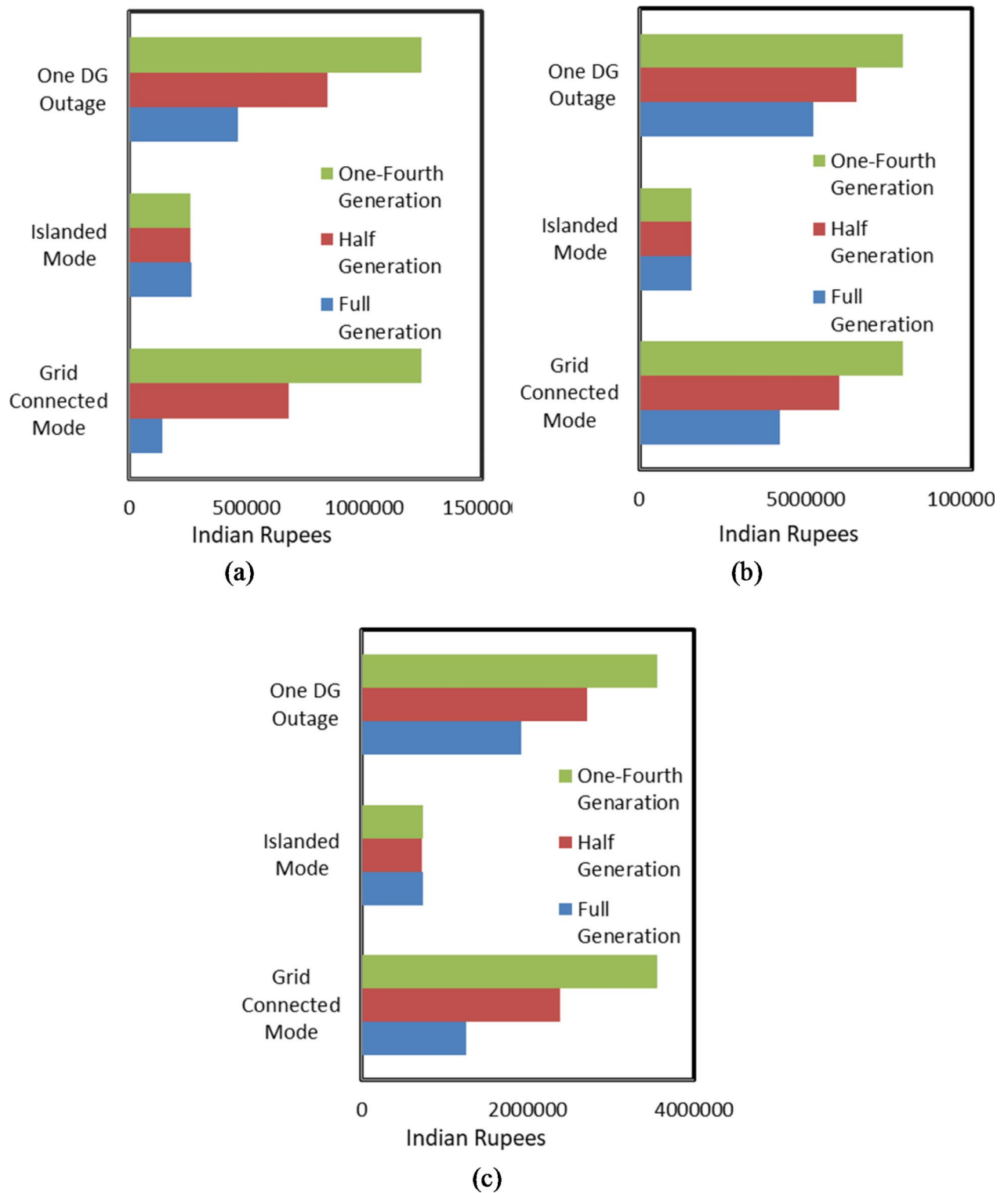


Fig. 9. TOC under different loading condition (a) light load (b) normal load (c) peak loa.

Voltage regulation (%)									
Case	Loading condition								
	Light load			Normal load			Peak load		
	FG	HG	OFG	FG	HG	OFG	FG	HG	OFG
Case 1	0.86	2.58	4	2.6	4.42	6.34	5.42	5.61	7.691
Case 2	0.1	1.18	6	0.23	0.07	0.82	0.43	0.37	0.98
Case 3	0.72	1.74	5	1.8	2.05	4.45	3.67	4.42	6.33

Table 4. Maximization of voltage regulation.

Case	Total loss reduction (%)	Cost savings (Rs.)	Voltage regulation (%)
Case 1	74.3	836,340	70.4
Case 2	92	1,359,240	97.2
Case 3	42.6	179,280	78.6

Table 5. Proposed approach results.

Optimization strategies	Total loss (kW)	Total operating cost (Rs)	Voltage regulation (%)	Computational time (s)
Interior point algorithm	197	2,273,811	4.3	124
Evolutionary algorithm	184	2,014,327	3.9	106
Proposed algorithm	93	1,263,660	2.6	34

Table 6. Comparison of the proposed technique to with other techniques.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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References

1. Yang, K., Liu, P. & J., & Relaying-assisted communications for demand response in smart grid: cost modeling, game strategies, and algorithms. *IEEE J. Sel. Areas Commun.* **38**(1), 48–60. <https://doi.org/10.1109/JSAC.2019.2951972> (2020).

2. Kiran, P. B. S. & Naran, P. Price setting of a microgrid operator in a radial distribution network. *IEEE Innov. Smart Grid Technol. - Asia (ISGT Asia)*. <https://doi.org/10.1109/ISGT-Asia.2019.8881697> (2019).

3. Wenlei, B., Ibrahim, E. & Kwang, Y. L. Optimal scheduling of distributed energy resources by modern heuristic optimization technique. *Int. Conf. Intell. Syst. Appl. Power Syst.* ; (2017). <https://doi.org/10.1109/ISAP.2017.8071407>

4. Pandya, S. B. et al. Multi-objective snow ablation optimization algorithm: an elementary vision for security-constrained optimal power flow problem incorporating wind energy source with FACTS devices. *Int. J. Comput. Intell. Syst.* **17**(1), 33 (2024).

5. Izci, D., Abualigah, L., Can, Ö., Andic, C. & Ekinci, S. Achieving improved stability for automatic voltage regulation with fractional-order PID plus double-derivative controller and mountain gazelle optimizer. *Int. J. Dyn. Control.* **1**, 1–6 (2024).

6. Premkumar, M. et al. Optimal operation and control of hybrid power systems with stochastic renewables and FACTS devices: an intelligent multi-objective optimization approach. *Alexandria Eng. J.* **93**, 90–113 (2024).

7. Pandya, S. B. et al. Multi-objective RIME algorithm-based techno economic analysis for security constraints load dispatch and power flow including uncertainties model of hybrid power systems. *Energy Rep.* **11**, 4423–4451 (2024).

8. Fadheel, B. A. et al. A hybrid sparrow search optimized fractional virtual inertia control for frequency regulation of multi-microgrid system. *IEEE Access.* (2024).

9. Manzoor, A. et al. AHHO: Arithmetic Harris Hawks Optimization algorithm for demand side management in smart grids. *Discover Internet Things.* **3**(1), 3 (2023).

10. Altawil, I. et al. Optimization of fractional order PI controller to regulate grid voltage connected photovoltaic system based on slap swarm algorithm. *Int. J. Power Electron. Drive Syst. (IJPEDS).* **14**, 1184–1200 (2023).

11. Altawil Ia, M. & Ka, Almomani, A. A grid connected hybrid renewable energy system for optimal energy managment based on ant-lion optimization algorithm. *J. Theor. Appl. Inform. Technol.* ;**101**(1). (2023).

12. Detroja, K. P. Optimal autonomous microgrid operation: a holistic view. *Appl. Energy.* **173**, 320–330 (2016).

13. Sarker, M. R., Ortega-Vazquez, M. A. & Kirschen, D. S. Optimal coordination and scheduling of demand response via monetary incentives. *IEEE Trans. Smart Grid.* **6**(3), 1341–1352 (2015).

14. Nan, S., Zhou, M. & Li, G. Optimal residential community demand response scheduling in smart grid. *Appl. Energy.* **210**, 1280–1289 (2018).

15. Zhang, Y., Gatsis, N. & Giannakis, G. B. Robust energy management for microgrids with high-penetration renewable. *IEEE Trans. Sustain. Energy.* **4** (4), 944–953 (2013).

16. Choudhury, S. et al. Energy management and power quality improvement of microgrid system through modified water wave optimization. *Energy Rep.* **9**, 6020–6041 (2023).

17. Khosravi, N. et al. A novel control approach to improve the stability of hybrid AC/DC microgrids. *Appl. Energy.* **344**, 121261 (2023).

18. Wei, W., Liu, F. & Mei, S. Energy pricing and dispatch for smart grid retailers under demand response and market price uncertainty. *IEEE Trans. Smart Grid*. **6**(3), 1364–1374 (2015).
19. Fu, Y., Shahidehpour, M. & Li, Z. Security-constrained unit commitment with ac constraints. *IEEE Trans. Power Syst.* **20**(3), 1538–1550 (2005).
20. Sahoo, G. et al. A novel prairie dog-based meta-heuristic optimization algorithm for improved control, better transient response, and power quality enhancement of hybrid microgrids. *Sensors* **23**(13), 5973 (2023).
21. Kahl, M., Freye, C. & Leibfried, T. A cooperative multi-area optimization with renewable generation and storage devices. *IEEE Trans. Power Syst.* **30**(5), 2386–2395 (2015).
22. Li, S., Zhou, J., Zhou, F., Niu, F. & Deng, W. A reduced current ripple overmodulation strategy for indirect matrix converter. *IEEE Trans. Industr. Electron.* 1–10. <https://doi.org/10.1109/TIE.2024.3453934> (2024).
23. Ouammi, D. H., Dessaint, L. & Sacile, R. Coordinated model predictive-based power flows control in a cooperative network of smart microgrids. *IEEE Trans. Smart Grid*. **6**(5), 2233–2244 (2015).
24. Conteh, F., Tobaru, S., Howlader, R., Yona, A. & Senjyua, T. Energy management systems for hybrid distributed generation sources in grid connected and stand-alone microgrids. *J. Renew. Sustain. Energy*. **9**, 065301 (2017).
25. Sahoo, G. et al. Scaled conjugate-artificial neural network-based novel framework for enhancing the power quality of grid-tied microgrid systems. *Alexandria Eng. J.* **80**, 520–541 (2023).
26. Chen, S. X. & Gooi, H. B. Jump and shift method for multi-objective optimization. *IEEE Trans. Industr. Electron.* **58**(10), 4538–4548 (2011).
27. Sampath Shirkhani, M. et al. A review on microgrid decentralized energy/voltage control structures and methods. *Energy Rep.* **10**, 368–380. <https://doi.org/10.1016/j.egy.2023.06.022> (2023).
28. Rezaei, N. & Kalantar, K. A novel hierarchical energy management of a renewable microgrid considering static and dynamic frequency. *J. Renew. Sustain. Energy*. **7**(033118), 1–20 (2015).
29. Behera, S. & Dev Choudhury, N. B. A systematic review of energy management system based on various adaptive controllers with optimization algorithm on a smart microgrid. *Int. Trans. Electr. Energy Syst.* <https://doi.org/10.1002/2050-7038.13132> (2021).
30. Behera, S. & Choudhury, D. N. B. Maiden performance analysis of PV and wind hybrid microgrid with battery management using PI and fuzzy controller connected with grid. In: Mohapatro, S., Kimball, J. (eds) Proceedings of symposium on power electronic and renewable energy systems control. Lecture Notes in Electrical Engineering, vol 616. Springer, Singapore. (2021). https://doi.org/10.1007/978-981-16-1978-6_33
31. Qinglin Meng, S. et al. Enhancing distribution system stability and efficiency through multi-power supply startup optimization for new energy integration. IET Generation Transmission & Distribution, early access. (2024). <https://doi.org/10.1049/gtd2.13299>
32. Tripathy, D. et al. A novel multiple-medium-AC-port power electronic transformer. *IEEE Trans. Industr. Electron.* **71**(7), 6568–6578. <https://doi.org/10.1109/TIE.2023.3301550> (2024).
33. Behera, S., Dev Choudhury, N. B. & Biswas, S. Maiden application of the slime mold algorithm for optimal operation of energy management on a microgrid considering demand response program. *SN Comput. Sci.* **4**, 491. <https://doi.org/10.1007/s42979-023-02011-9> (2023).
34. Marzband et al. Experimental validation of a real-time energy management system using multi-period gravitational search algorithm for microgrids. *Appl. Energy*. **128**, 164–174 (2014).
35. Li, P., Hu, J., Qiu, L., Zhao, Y. & Ghosh, B. K. A distributed economic dispatch strategy for power–water networks. *IEEE Trans. Control Netw. Syst.* **9**(1), 356–366. <https://doi.org/10.1109/TCNS.2021.3104103> (2022).
36. Roy, K., Mandal, K. K. & Mandal, A. C. Modeling and managing of micro grid connected system using Improved Artificial Bee colony algorithm. *Int. J. Electr. Power Energy Syst.* **75**, 50–58 (2016).
37. Abdolrasol, M. G. et al. An optimal scheduling controller for virtual power plant and microgrid integration using the binary backtracking search algorithm. *IEEE Trans. Ind. Appl.* **54**(3), 2834–2844 (2018).
38. Witharama, W. M. N., Bandara, K. M. D. P. & Azeez, M. I. Kasun Bandara; V. Logeeshan; Chathura Wanigasekara, Advanced genetic algorithm for optimal microgrid scheduling considering solar and load forecasting, battery degradation, and demand response dynamics. *IEEE Access*. **12**, 83269–83284 (2022).
39. Khadanga, R. K., Padhy, S., Panda, S. & Kumar, A. Design and analysis of multi-stage PID controller for frequency control in an islanded microgrid using a novel hybrid whale optimization-pattern search algorithm. *Int. J. Numer. Model.* e2349. <https://doi.org/10.1002/jnm.2349> (2018).
40. Sumit Sharma, Y. R. et al. Salah Kamel, modeling and sensitivity analysis of grid-connected hybrid green microgrid system. *Ain Shams Eng. J.* **13**(4). <https://doi.org/10.1016/j.asej.2021.101679> (2022).
41. Dashtdar, M., Bajaj, M., Hosseinimoghadam, S. M. Design of optimal energy management system in a residential microgrid based on smart control. *Smart Sci.* <https://doi.org/10.1080/23080477.2021.1949882> (2021).
42. Amin, K. Microgrid optimal scheduling with multi-period islanding constraints. *IEEE Trans. Power Syst.* **29**(3), 1383–1392 (2014).
43. Sharma, S., Sood, Y. R., Kumar, V., Sharma, N. K., Bajaj, M., Jurado, F., Kamel, S. Optimal sizing and cost assessment of off grid connected hybrid microgrid system IEEE Global Power, Energy and Communication Conference - GPECOM Cappadocia, Turkey June 14–17, pp. 344–348, 2022. (2022). <https://doi.org/10.1109/GPECOM55404.2022.9815817>
44. Tamil nadu electricity regulatory commission. Suo Motu determination of tariff for generation and distribution, Order dated (11-12-2014); 6.12–6.15.
45. Selvaraj, G. & Rajangam, K. Multi-objective grey wolf optimizer algorithm for combination of network reconfiguration and D-STATCOM allocation in distribution system. *Int. Trans. Electr. Energy Syst.* e12100. <https://doi.org/10.1002/2050-7038.12100> (2019).
46. Ramya, R., Sivakuaran, T. S. An efficient RFCSA control strategy for PV connected quasi Z-source cascaded multilevel inverter (QZS-CMI) system. *Int. J. Numer. Model.* **33**, e2660. <https://doi.org/10.1002/jnm.2660> (2020).
47. Askarzadeh, A. A novel metaheuristic method for solving constrained engineering optimization problems: crow search algorithm. *Comput. Struct.* **169**, 1–12 (2016).

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Declarations

Competing interests

The authors declare no competing interests.

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